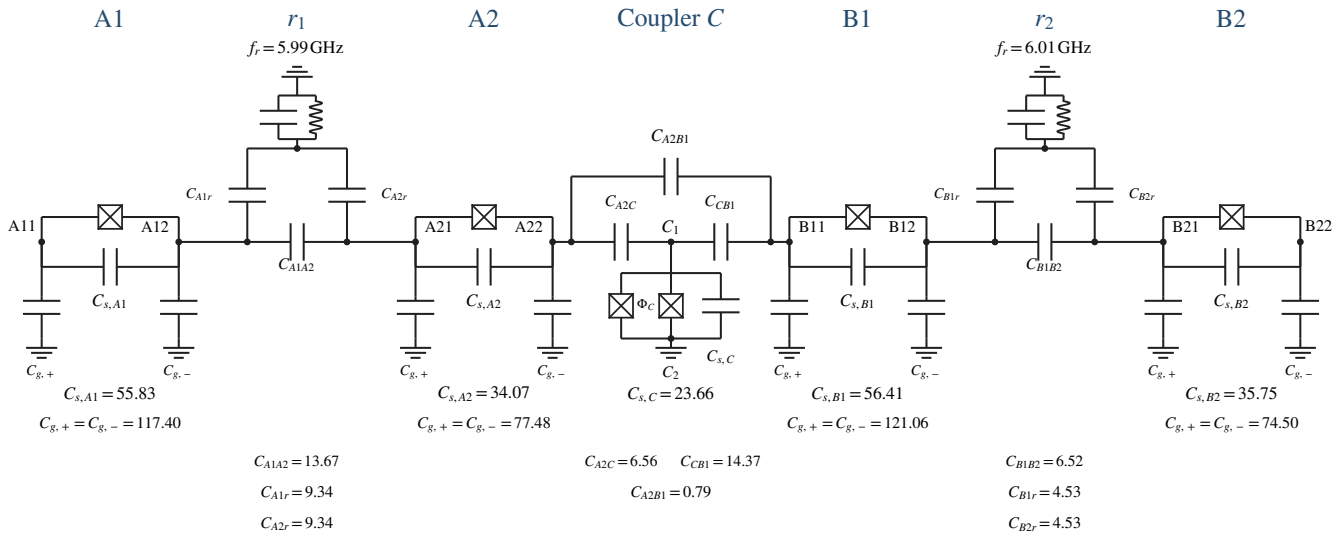
1 | Circuit and rail SQUID design



All capacitances are in fF. Readout branches are contextual; decoherence calculations exclude resonator dynamics.

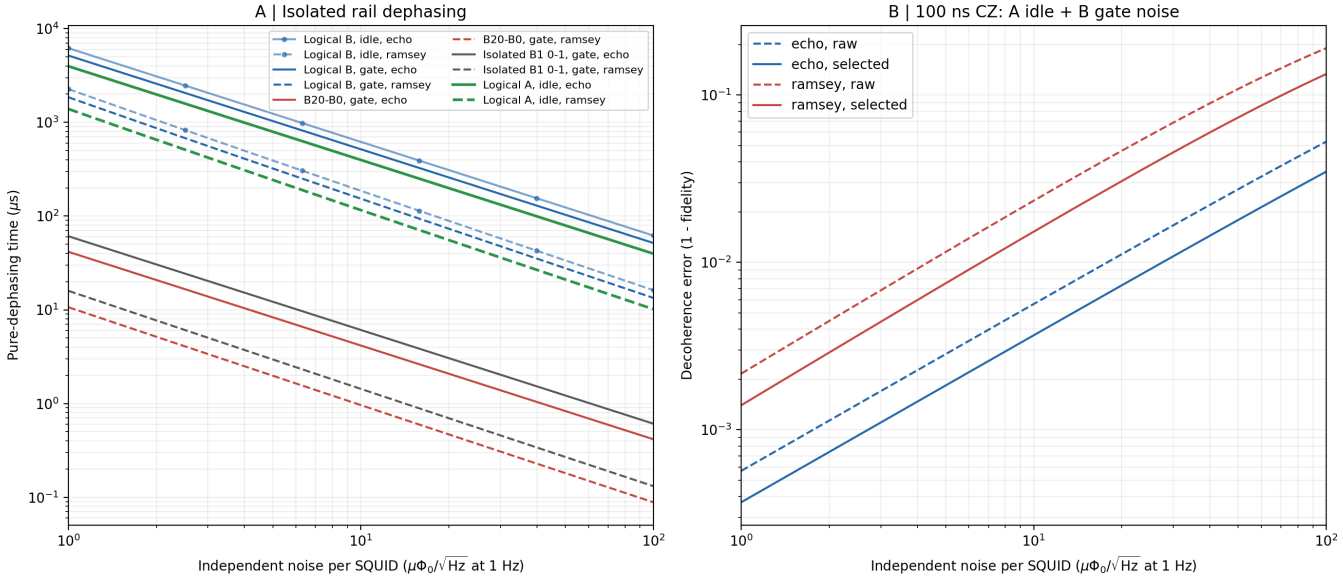
Circuit context: B-noise omits C and resonators; C-noise uses one pair plus C. Rail JJ symbols represent effective SQUIDs. Loaded EC and EJ values are collected below; capacitance values remain on the drawing.

Transmon	Loaded EC/h (MHz)	Bare f01 range (GHz)	Large junction EJ/h (GHz)	Small junction EJ/h (GHz)	Asymmetry d	Effective EJ/h idle/OFF / gate/ON (GHz)
A1	165.00	4.00 - 5.00	16.72	3.53	0.65	17.28 / 17.28
A2	250.00	4.00 - 5.00	11.48	2.37	0.66	11.91 / 11.91
C	450.28	6.50 - 11.00	25.03	11.49	0.37	29.30 / 15.75
B1	160.00	4.40 - 5.80	22.04	5.75	0.59	24.26 / 19.96
B2	260.00	4.40 - 5.80	14.12	3.61	0.59	15.33 / 12.65

Asymmetric SQUIDs, negligible loop inductance: EJ,max = EJ,large + EJ,small; EJ,min = EJ,large - EJ,small; d = EJ,min/EJ,max. Junction energies use the numerical full-cosine f01 at the specified loaded EC. A tunability is a hardware specification; the central CZ estimate includes A at idle and B at gate.

Input	Value
B idle / gate frequency (GHz)	5.39000 / 4.87128
Ideal gate duration (ns)	100.0
1/f cutoff assumptions	Ramsey low: 1 Hz; no finite high cutoff
Local transmon levels	7

2 | Isolated rails and CZ decoherence error



Panel A: logical A at idle; logical B at idle/gate; B20-B0 and isolated B1 only at gate. Panel B: log-scale error for the ideal 100 ns B20-inclusive CZ, with independent A-idle and B-gate noise. The auxiliary has A in its pair ground. No coupler noise is added.

Noise per loop	Echo raw	Echo selected	Ramsey raw	Ramsey selected
1	99.9432%	99.9631%	99.7835%	99.8599%
10	99.4358%	99.6330%	97.6620%	98.4734%
100	94.7329%	96.5134%	80.9276%	86.6408%

First-order echo/Ramsey references substituted into exponential pure dephasing. Independent A-idle and B-gate generators added; correlations among logical and auxiliary states retained. B20 has A in pair ground. Equal noise amplitude per physical SQUID. No coupler noise, T1, quadratic/transverse noise or ramps in the central gate estimate.

3 | Isolated B-rail details

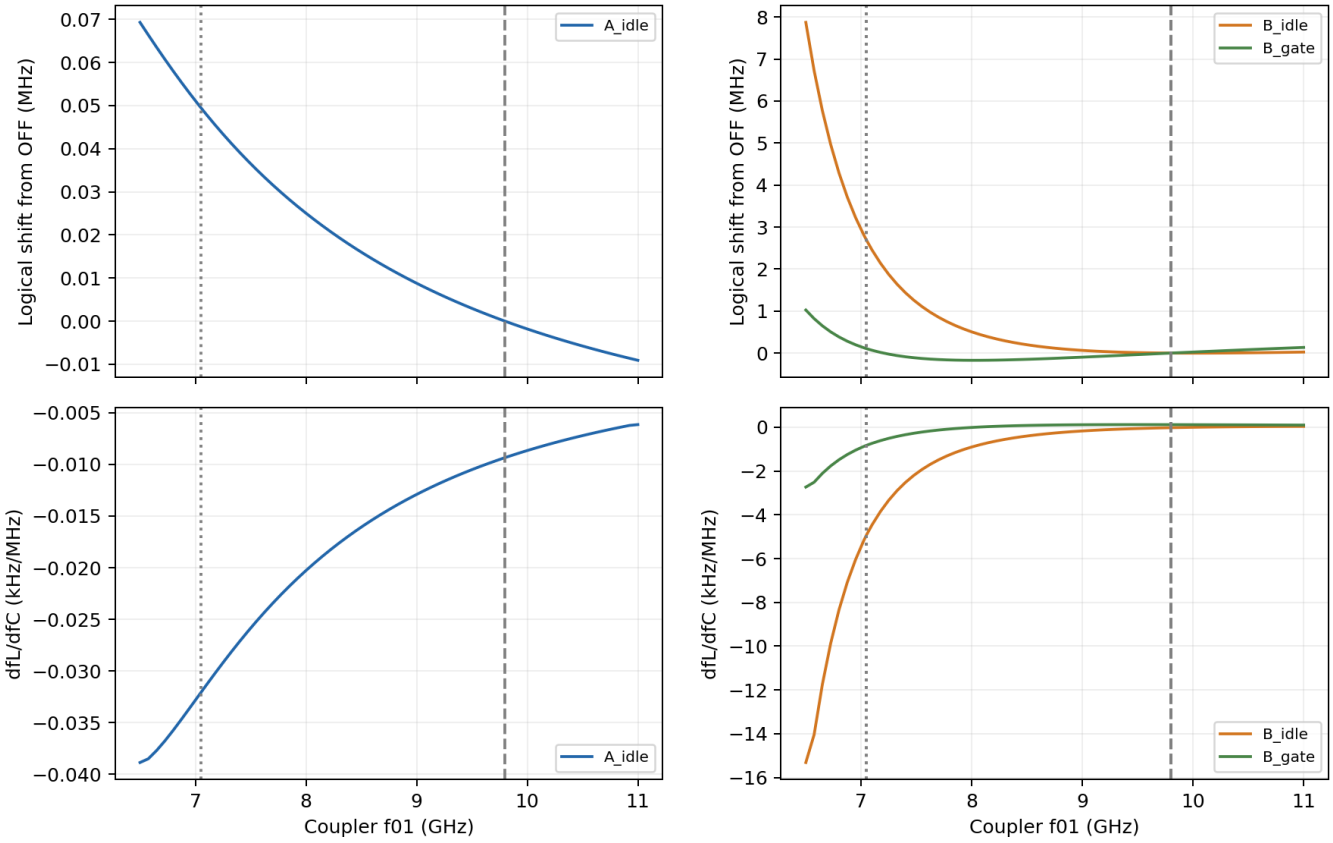
SQUID parameter	B1	B2
Loaded EC/h (MHz)	160.00	260.00
Large junction EJ/h (GHz)	22.04	14.12
Small junction EJ/h (GHz)	5.75	3.61
Asymmetry	0.5861	0.5932

Position	Coherence	Echo (us)	Ramsey (us)
idle	Logical B	619	186
idle	B20-B0	5.09	1.19
idle	Isolated B1 0-1	7.32	1.73
gate	Logical B	516	153
gate	B20-B0	4.16	0.962
gate	Isolated B1 0-1	6.09	1.43

Times use the sampled amplitude nearest10 microPhi0/sqrt(Hz) at1Hz. The isolated B1 reference removes B2 and C at the same B1 frequency and loaded EC. B20-B0 is the auxiliary phase coherence across the B component of the selected logical10/B20 transition, not a population lifetime.

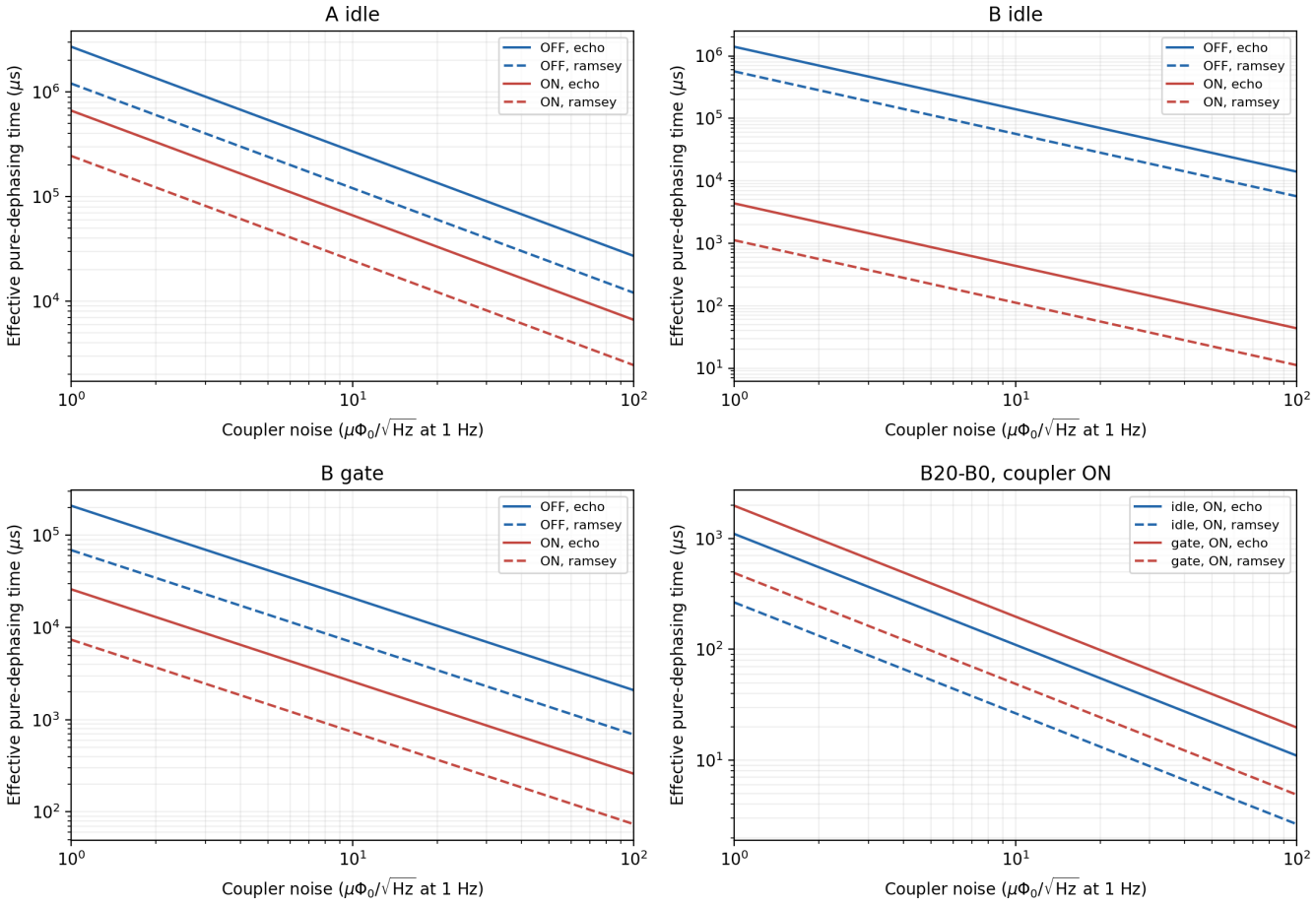
Both bare B frequencies are calibrated to resonance independently at idle and gate. Differential detuning is first-order protected, but flux dependence of charge matrix elements changes the pair coupling. This calculation includes first-order longitudinal sensitivity only; quadratic and transverse noise are excluded.

4 | Separate coupler-induced dephasing



Rail EJs are noiseless. Differential rail bias is calibrated once at idle with C OFF, then held fixed for the common move to gate. Only C EJ changes along the sweep. Dashed/dotted lines denote OFF/ON. This differs from independently resonating the isolated B pair at every operating point; the comparison is not an identical-control subtraction.

5 | Coupler-induced dephasing times



Times are in microseconds. Reference amplitude100 microPhi0/sqrt(Hz) at1Hz is scaled inversely with amplitude for the adopted exponential-rate convention. The Ramsey cutoff logarithm is held at its reference value in this separate C-noise figure; isolated B Ramsey times are solved independently at every amplitude. B20 is shown only relative to B0 and only with C ON. No intrinsic rail noise or T1 is added here.

6 | Relative size, interpretation and checks

Gate coherence	Rail Tphi (us)	C Tphi (us)	Rail rate / C rate
Logical B	51.61	259.58	5.03
B20-B0	0.42	19.74	47.49

Compare the listed rate ratios rather than assume a uniform order-of-magnitude reduction. Static dressing and different calibration conventions can matter separately from adding a noise source. Combined-noise decoherence and fidelity are intentionally omitted from this report.

Gate rates are phenomenological exponential substitutions of echo/Ramsey first-order 1/f reference times. Independent A/B generators add, retaining state correlations and the A-ground/B20 auxiliary. Rate matrices are checked for complete positivity. No T1, thermal population, gate-matrix-element noise, pulse ramps or baseline coherent error is included. Separate coupler-noise curves remain diagnostics, not an added gate-error source here. All inputs and derivatives remain in results.json.

7 | A-rail tunability comparison (alternative only)

The nominal A1/A2 design remains 4.00-5.00 GHz (1 GHz span), as used on pages 1-6. The alternative is 4.36-4.86 GHz (500 MHz total span), centered at the same 4.61 GHz idle setting. This comparison both narrows the range and recenters it. Loaded ECs, idle position, B design and the 100 ns exponential gate model are unchanged. No coupler noise is added.

A transmon	Large EJ/h: nominal / alternative (GHz)	Small EJ/h: nominal / alternative (GHz)
A1	16.72 / 17.37	3.53 / 1.81
A2	11.48 / 11.91	2.37 / 1.21

Isolated logical A at idle: dephasing times in microseconds. Noise amplitude is in $\mu\Phi_0/\sqrt{\text{Hz}}$ at 1 Hz per independent physical loop.

Noise	Echo: 1 GHz	Echo: 500 MHz	Ramsey: 1 GHz	Ramsey: 500 MHz
1	3969.08	7476.62	1388.42	2793.45
10	396.91	747.66	115.76	227.71
100	39.69	74.77	10.16	19.77

Full B20-inclusive CZ with A-idle and B-gate noise: fidelities below are percentages, corresponding to error 1-F in page 2 panel B.

Noise / rates	Raw: 1 GHz	Raw: 500 MHz	Selected: 1 GHz	Selected: 500 MHz
1 / echo	99.9432%	99.9499%	99.9631%	99.9675%
1 / ramsey	99.7835%	99.8087%	99.8599%	99.8763%
10 / echo	99.4358%	99.5017%	99.6330%	99.6764%
10 / ramsey	97.6620%	97.9272%	98.4734%	98.6493%
100 / echo	94.7329%	95.3132%	96.5134%	96.9087%
100 / ramsey	80.9276%	82.5976%	86.6408%	87.9198%

The narrower-range alternative approximately doubles logical A coherence. At noise amplitude 10, total raw and selected gate errors decrease by about 12%; B noise still contributes most of the error. This alternative is not adopted as the nominal design. Complete comparison values are retained in tuning-comparison.json.